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Translating artificial intelligence into socio-economic insight: a hybrid deep learning approach to employee financial well-being

Aakanksha Uppal[1*], Anubha Srivastava[2], Yashmita Awasthi[3], Anjita Srivastava[4] and Barkha Kakkar[2]

*Correspondence:
Aakanksha Uppal
aakanksha.uppal@symlaw.edu.in
[1]Symbiosis International (Deemed University) Pune, Symbiosis Law School Noida Campus, Noida, Ghaziabad, India
[2]Institute of Technology & Science, Mohan Nagar, Ghaziabad, India
[3]School of Commerce, Finance and Accountancy, Christ University, Bengaluru, India
[4]Bundelkhand University, Jhansi, India

Abstract
This study aims to translate recent advancements in hybrid artificial intelligence (AI) modeling into a functional tool for assessing individual financial well-being. The objective is to develop a system that aids organizations in understanding employees' financial stress, with broader implications for enhancing workplace productivity and societal economic resilience. A deep learning pipeline was developed to classify individuals into three financial well-being categories: *Financially Secure*, *Moderately Stable*, and *Financially At-Risk*. The approach utilizes a structured dataset of 20,000 Indian individuals and implements 15 advanced deep learning models, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Gated Recurrent Units (GRU), Bidirectional Long Short-Term Memory (BiLSTM), and Wide & Deep networks. Model performance was assessed using standard evaluation metrics, including validation accuracy and ROC-AUC scores. Among the tested models, the hybrid *Wide & Deep + CNN* configuration yielded the highest performance, achieving a validation accuracy of 99.44% and a perfect ROC-AUC score of 1.0000. These results validate the model's capacity for robust classification and real-world applicability to financial profiling. This study demonstrates a practical application of AI in financial decision support systems and contributes to organizational research by offering a scalable solution to assess and mitigate employee financial stress.

Keywords Financial well-being assessment, Deep learning, Wide & deep network, CNN, Employee financial stress, Organizational productivity, Societal economic stability

1 Introduction

In the context of organizational dynamics, financial decision-making is not merely a technical or administrative function, but a reflection of deeper organizational structures, behaviors, and societal interactions. The journal underscores the need to explore how financial practices influence and are influenced by the lived experiences of individuals within organizations. This encourages studies that move beyond abstract theorizing, focusing instead on the transformative application of theoretical insights into real-world organizational practices. Thus, integrating financial analysis with organizational theory contributes not only to scholarly understanding, but also to the practical improvement of life within and across institutions.

Finance refers to the systematic management, investment, and allocation of money to preserve its value, build it over time, or simply for efficient utilization. It is a critical tool in the decision-making process of an individual, an organization, or a government. Finance has generally been divided into three major fields: public finance, corporate finance, and personal finance [24]. Public finance deals with a nation's revenues and expenditures including tax policies, budgeting, public debt, and economic stabilizing efforts. The idea of corporate finance reveals how a corporation may manage its financial actions with regard to the acquisition of assets, liabilities, investments, and the raising of capital through debt to ensure profitability and sustainability. Personal finance refers to money management by an individual or family, including budgeting and saving, investing, insurance planning, mortgage management, and retirement planning [4, 25].

At its core, the concept of financial literacy focuses on how to manage money wisely, which can be encapsulated by two key principles saving and investing. An essential rule is "pay yourself first," which acknowledges putting your financial well-being left first saving and investing prior to spending on any frivolities. An extension of this line of thought is investment in oneself through education or furthering skill development, which is arguably a key step toward long-term financial independence [8]. Equally important is being wary of bad debt-typically, a concept borrowed for non-productive purposes that offers no possibility of return, such as impulsive purchasing of luxury items. Conversely, a debit for education or even starting a business may pay off in the near or distant future. An emergency fund is another must-have item to stay cushioned by the financial shocks arising from unexpected events. Thus, getting our personal finance in shape through wise decisions brings about security, less stress, and a peaceful and stable life [14].

The graph "Financial Security and Stress Indicators (2025)," as shown in Fig. 1, provides a comparative overview of the key financial well-being indicators. It divides the world from U.S. survey data. The emphasis is on at mere 29% of the global population that feels optimistic about its financial future, indicating a sharp decline in the public's confidence in recent times due to inflation, rising costs of living, and economic

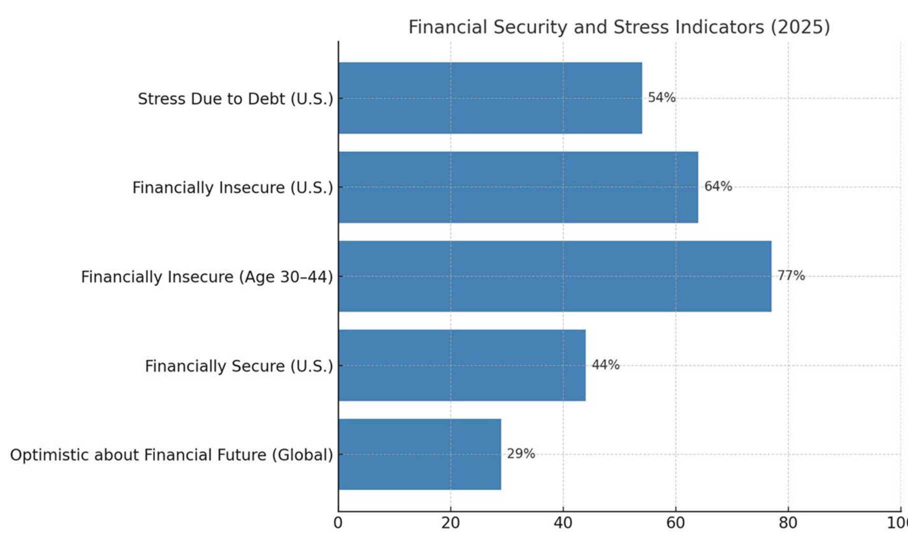

Fig. 1 Financial security and stress indicators [9]

instability. In the United States, 44% of the population reports that they feel financially secure from the increase from the previous year, yet the majority of the people seem to be fighting some level of financial strain. The age bracket between 30 and 44 years has been cited as particularly at risk, with 77% of adults feeling financially insecure. This group generally encounters major expenses, such as student loans, mortgage payments, and child expenses, thus placing them under enormous economic pressure. There are 64% of Americans feel financially insecure; this, implies systemic challenges faced in achieving economic stability, even in a high-income economy. In addition, 54% of adults in the U.S. claimed to feel stressed because of indebtedness, which significantly affects mental health and overall well-being [9].

This visualization again speaks of continuous financial insecurity and calls for urgent interventions such as debt management, financial literacy, and policy reforms that will build long-term economic resilience, particularly for the working population.

The conventional approach to measuring financial well-being involves manual analysis of records of income and expenditure, credit scores, budget worksheets, or self-reported surveys. Financial institutions and policymakers rely on static indicators to measure individuals' and households' financial health. Such static markers may include debt-to-income ratios, savings rates, or credit utilization. These indicators can provide a general measure, but often fail to represent true financial behavior, which is both dynamic and nonlinear [12]. On the contrary, they are reactive, mostly descriptive rather than predictive, lack personal customization, and do not accommodate the real-time variability exhibited in financial patterns. Their set limitations include an inability to scale efficiently, a high degree of susceptibility to reporting errors, dependence on lengthy data collection procedures, and an inability to detect intricate behavioral patterns, such as impulsive spending or irregular income streams—behaviors fairly common among gig workers [20].

Artificial Intelligence (AI) and Hybrid Deep Learning models overcome the limitations that hinder real-time data-driven financial well-being assessments. These models handle massive amounts of structured and unstructured financial data, such as transaction logs, digital receipts, behavioral trends, and lifestyle patterns. Hybrid models combine CNNs Convolutional Neural Networks (CNN) for feature extraction with RNNs Recurrent Neural Networks (RNN) for sequence modeling to capture complex temporal patterns in income and expenditure over time. These can identify early warning signs of an impending financial freeze, predict the future financial state, and offer customized financial relief. Furthermore, AI systems adapt and learn with the arrival of each set of new data, thereby increasing their accuracy and relevance. With this, they provide far broader, predictive, and scalable approaches to assessing financial well-being compared to conventional approaches [5, 15, 17].

1.1 Contribution

The following points summarize the primary contributions of this study, focusing on the core innovations and technical achievements that define its impact and significance.

1. A comprehensive dataset of 20,000 Indian individuals was developed, incorporating detailed financial information, such as potential savings in nine specific categories, desired savings percentage, debt ratio, occupation, and city tier.

2. Engineered new features such as the *Savings Ratio* (Desired Savings ÷ Income) and *Debt Ratio* (Loan Repayment ÷ Income), were shown to significantly influence model predictions, accounting for up to 50% of decision-making in the TabNet model. These types of features are rarely used in the existing financial profiling studies. Designed a unique hybrid machine learning model combining Wide & Deep learning with Convolutional Neural Networks (CNNs), even though the data were non-sequential. This approach treats financial records as sequences by reshaping the data, thereby allowing CNNs to learn complex patterns.
3. Two types of correlation heatmaps were used one for the original data and the anotherfor savings-related features to uncover hidden financial behaviors. For example, astrong correlation (r = 0.89) was found between grocery expenses and grocery savingspotential, guiding both the feature selection and model interpretation.
4. TabNet was applied to understand which features the model relied on the most. Surprisingly, traditional metrics such as income and age were less important thanbehavioral indicators such as debt ratio and miscellaneous savings potential. Thisinsight challenges existing financial modeling practices and emphasizes the value ofbehavior-based features.

2 Related work

In recent years, a growing body of interdisciplinary research has sought to unravel the multifaceted dimensions of financial well-being and systemic stability, using advanced computational methods, machine learning, and behavioral modeling. The collective objective is to understand financial stress better, predict financial risks, and build resilient systems that bridge the gap between individual experiences and broader economic phenomena.

Ghashti et al. [11] conducted a foundational study in this direction with the intention of identifying the major sources of financial stress by analyzing the responses of 1,874 individuals to a set of 68 mixed-type survey questions collected in 2022. Distance-based clustering an effective tool in the toolkit of any financial segmented, was employed with a mixed-type distance incorporating variable-specific kernel functions with cross-validated bandwidths in order to discriminate between relevant and irrelevant variables effectively. The analysis yielded two major clusters: steady savers, indicating strong financial well-being and financial strivers, comprising individuals undergoing high financial stress. The segmentation yielded actionable insights into personal financial advices and paved the way for automated financial advising and investment generation. While individual stress is critical, Polyzos et al. [22] expanded the lens to systemic shocks, such as banking crises, affecting subjective well-being. By setting an agent-based modeling framework alongside a support vector machine (SVM) subjective well-being function, researchers simulated the direct and indirect effects of economic downturns, such as income loss, unemployment, and psychological distress. However, the associated research findings, show that welfare losses from bank failures often exceed the fiscal cost of government bailouts. They further underscored the asymmetry in losses among various segments of the population, thus revealing the social complexity underlying economic policymaking. Gensler & Bailey [10] focused on the emerging systemic risks brought about by the widespread adoption of deep learning-based systems in financial ecosystems, mapping this impact across five transmission pathways. A paradox was brought out: while these models provide stellar levels of prediction and efficient performance, they simultaneously increase interconnectedness within the systems and may even lend new forms of fragility to these systems. The study concluded that traditional regulatory frameworks are insufficient for the evolving challenges, and it proposed that policy tools should be reconsidered to protect the stability of the financial system in the AI age. To complement this, Alessi and Savona [2] addressed the technical limitations of conventional financial risk models. They state that traditional empirical methods are incapable of representing the complex, nonlinear, and multidimensional nature of financial data. The imbalanced and high-dimensional nature of datasets makes them an ideal domain for machine learning, which provides a more robust solution. However, despite being extraordinarily good in terms of predictive accuracy, such models, which are largely taken to be black boxes, ironically remain the least used in the assessments of mainstream financial stability, primarily due to their perceived lack of interpretability, thus signaling the gap between innovation and regulatory acceptance. To address these issues, Khunger [16] proposed a deep learning-based framework for financial stress testing Their architecture, Correlation CNN-LSTM, combines both quantitative and qualitative indicators. The model was highly accurate with an almost negligible training loss (0.0013) and testing loss (0.003). It predicts core financial variables fairly well. The model was able to decrease major financial risk types, such as credit, liquidity, market, and operational risks, to a great extent, thereby furthering the usefulness of financial risk measurements and systemic resilience. Building on this, Shi et al. [23] developed the Hybrid Financial Risk Predictor (HFRP), which is a model similar to the CNN and LSTM architectures but is especially useful in combining numerical data with financial text analytics. Similar to the earlier setup, it achieved very high accuracy and stability with training and testing losses again posted at 0.0013 and 0.003, respectively. It provided accurate forecasts for important variables, such as revenue, net income, and EPS, while also mitigating different categories of risks. The hybrid nature of the model shows how combining data modalities can enrich the provisions of risk information, that can be used to make dependable financial decisions. Further refinement of model selection led to the development by Elhoseny et al. [7] of a novel Adaptive Whale Optimization Algorithm with Deep Learning (AWOA-DL) for the prediction of financial distress. The three-step pipeline, namely data preprocessing, hyperparameter optimization based on AWOA, and prediction using a multilayer perceptron-based deep neural network, was validated on four datasets. With an average accuracy of 95.8%, the models were found to be superior to the classical methods, with corresponding benchmark accuracies of 93.8%, 89.6%, 84.5%, and 78.2%. This study went a long way in supports the notion that evolutionary optimization techniques can significantly improve the accuracy and robustness of deep learning models in finance. In continuation with predictive modeling for distress scenarios, Naved et al. [21] dealt with bankruptcy prediction aided by machine learning and metaheuristic optimization. This research applied the Histogram Gradient Boosting Classification (HGBC) model and hybridized it with three optimization algorithms: Snake Optimization Algorithm (SOA), Gradient-Based Optimization (GBO), and Bonobo Optimization Algorithm (BOA), resulting in the three hybridized models HGSO, HGGB, and HGBO, respectively. With pure HGBC, precision was 0.940, while HGBO and HGGB increased it to 0.950 and 0.960, respectively. However, the HGSO model stands out with the highest precision level of 0.980, establishing it as an

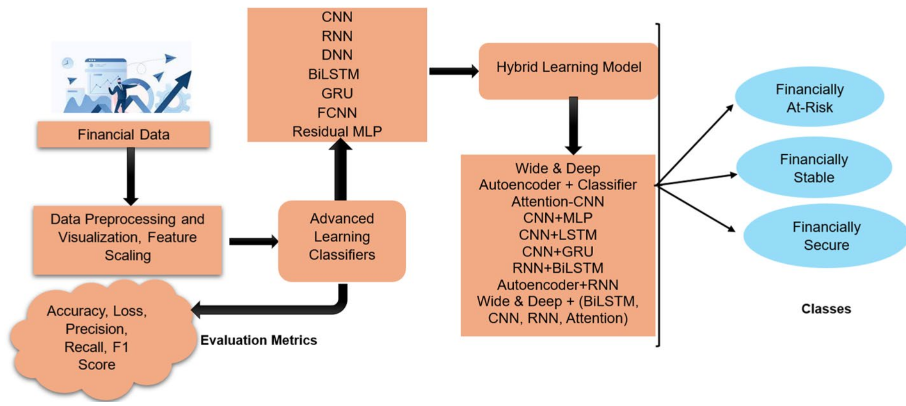

Fig. 2 Proposed system to detect and classify the financial status

Table 1 Categorization and description of features used for financial well-being analysis

Category	Feature	Description
Income & demographics	Income	Monthly income of the individual
	Age	Age in years
	Occupation	Type of employment or job role
Monthly expenses	Rent	Monthly rent payments
	Loan_Repayment	Monthly loan repayment obligations
	Insurance	Insurance premiums paid monthly
Financial goals & savings	Desired_Savings	Targeted monthly savings amount
	Disposable_Income	Remaining income after deducting all expenses

elite early bankruptcy detector and holds great promise for the preemptive mitigation of financial risk. Finally, Akash et al. [1] study the relationship between systemic risk and financial indicators through a regression study of Yahoo stock market data. With liquidity, solvency, profitability, and risk exposure, a strong positive correlation is found, mainly between liquidity and profitability, indicating that firms with a high liquidity level tend to perform well financially. The regression also shows that as much as 94% of the variance in profitability can be predicted from these indicators, and price-earnings (PE) ratios are affected mostly by stock price trends. They further emphasized that comprehensive financial analysis ensures risk management for long-term financial stability.

3 Proposed pipeline to analyse the financial status

The pipeline proposed in this section helps understand and classify people based on their financial behavior. As shown in Fig. 2, different deep learning models are applied to the data to learn higher-level abstractions. To further increase performance, hybrid models are established by two or several existing neural architectures and finally the system assigns individuals to the three major financial states of Financially At-Risk, Financially Stable, and Financially Secure.

3.1 Financial dataset description

The data contains detailed information on the financial and demographic aspects of 20,000 individuals from India. It includes details on monthly income, expenditure patterns, goals for savings, and potentials savings in categories such as groceries, transport, and healthcare as shown in Table 1. This rich dataset is well suited for personal finance behaviors, cost-optimization opportunities, and predictive model developments in financial planning. As a wider snapshot of expenditures and their present saving capacity, the dataset may act as an important resource for researchers, data scientists, and financial analysts interested in the Indian financial industry [13].

Additional supporting features include Dependents, which signify the number of people financially supported by the person, and City_Tier, which corresponds to regional economic disparities by assigning cities into tiers. Expenses from the dataset were finely detailed and split by month for various categories, including Groceries, Transport, Eating_Out, Entertainment, Utilities, Healthcare, Education, and Miscellaneous. These are crucial for estimating Potential Savings, which is an assessment of opportunities to reduce spending and thereby improve financial outcomes. Furthermore, Desired_Savings_Percentage functions as a subjective metric for financial goal setting, thereby assisting in measuring the gap between the desired and actual saving behavior. Together, these variables support the study of financial wellness and spending patterns fairly thoroughly.

3.2 Data preprocessing and visualization of financial data

Several preprocessing steps are required to effectively prepare dataset for machine-learning models. First, categorical features such as "Occupation" and "City_Tier" are Label Encoded so as to convert categories represented by textual characters into numerical values without losing relational semantics, thus allowing the algorithms to process them. Subsequently, feature engineering is applied to create new meaningful variables, such as the Savings Ratio (Desired_Savings divided by Income) and Debt Ratio (Loan Repayment divided by Income). These ratios provide normalized insight into personal saving behavior and financial liability, so that the model reviews personal financial health across various income levels. Finally, standardization was applied to all numerical features to rescale the data to have a mean of 0 and a standard deviation of 1. In this way, larger measurements do not dominate smaller ones and help in faster convergence and more stable learning using gradient-based algorithms. These steps boost the goodness of prediction for the dataset thus, models can learn more efficiently and accurately.

As presented in Fig. 3, the graph titled Income Distribution with Mean and Median provides a visual perception of how the incomes of individuals are distributed within this particular dataset. The distribution is highly right-skewed; this means that, while most individuals earn lower and moderate incomes, a small number of people earn very high incomes, pulling the tail of the distribution to the right. This is further validated by the fact that the mean income (41,585) is above the median (30,185), which is a well-known feature of skewed data. The importance of this graph lies in its implications for the statistical analysis and model building. In a skewed dataset, such as this, the median often acts as a better measure of central tendency than the mean and is especially used for describing the income of a typical individual. In addition, this skewness advocates employing normalization techniques (for example, log transformation) to income values before feeding such features to machine-learning algorithms to reduce bias and improve convergence. The application of this skew can result in different feature engineering strategies, such as Income brackets or percentiles, for example, allowing models to better distinguish between low-, middle-, and high-income groups. Ultimately, the graph conveys that income data needs to be treated with care to avoid distortions in income-based predictions and ensure that models learn in a fair and balanced manner.

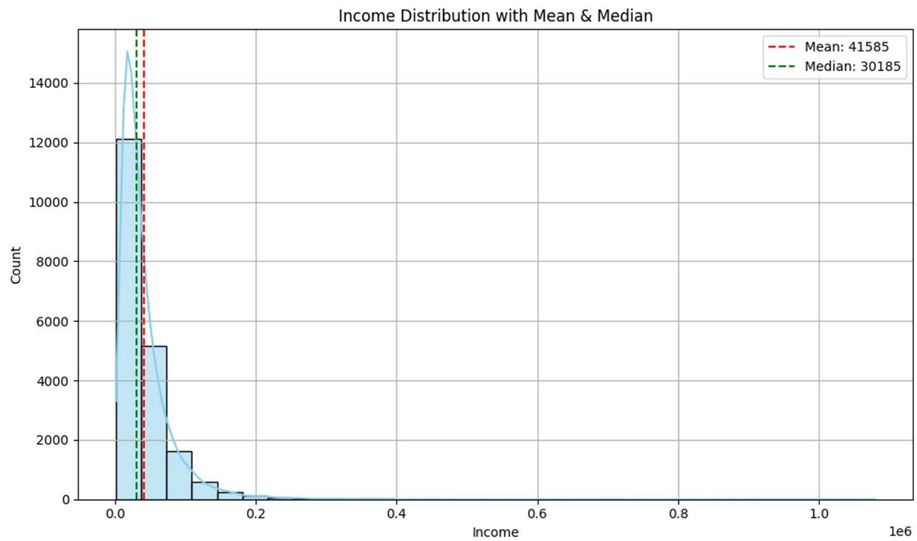

Fig. 3 Income distribution with mean & median

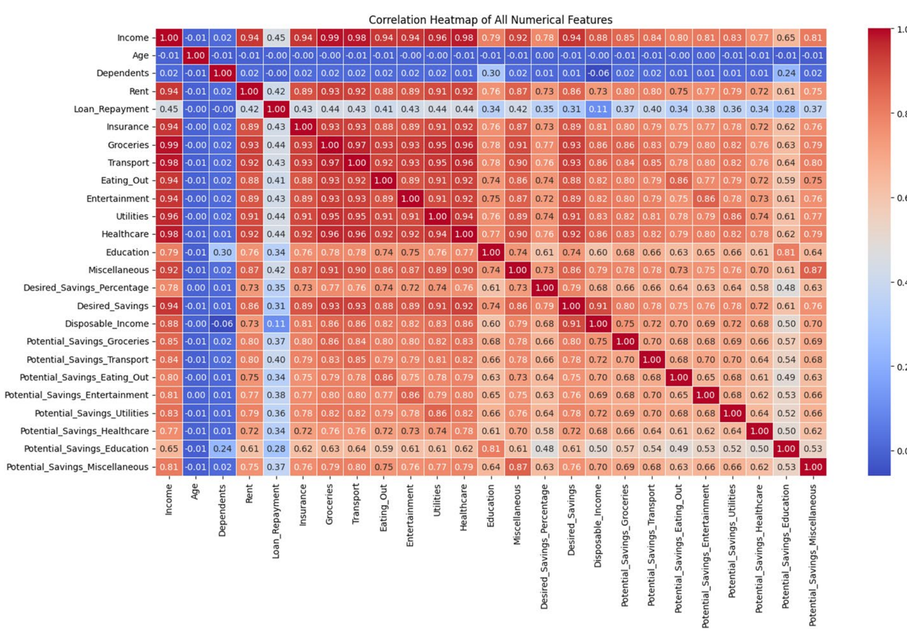

Fig. 4 Correlation heatmap of numerical features

The correlation heatmap in Fig. 4 provides a detailed view of the interrelationships among all the numerical features in the personal finance dataset. One of the key observations is the positive correlation of Income with the following expense categories: groceries (0.99), insurance (0.94), education (0.79), and healthcare (0.94). This means that income goes up, and so does expenditure of the individuals across these categories-a proportional spending style. There are high positive correlations between Potential Savings and Income across all categories (groceries, utilities, education, etc.), as well as the actual expenditures for these categories (≥ 0.75). For example, Potential_Savings_Groceries correlates at 0.89, and Potential_Savings_Utilities at 0.82, with actual spending, meaning that potential savings are calculated as fraction of current spending, which is correlated with income. Similarly, Disposable_Income positively correlates with desire savings (0.86) and income (0.88), which is logical: those with higher residual income after expenses tend to aim for higher savings. Desired_Savings_Percentage, however, moderately correlates with Income (0.35), and hence, saving intent (as a proportion of income) varies independently of actual income levels and probably reflects behavior or attitudes. The heatmap also reveals clusters that comprise highly correlated features, such as expenses (Groceries, Transport, Healthcare, etc.) and Potential Savings variables. Such clusters could imply multicollinearity, which is important to consider when making decisions about feature selection or dimensionality reduction in predictive modeling.

The potential Savings correlation heatmap in Fig. 5 shows moderate to strong positive correlations across categories, mostly ranging from 0.60 to 0.70. For example, Potential_Savings_Groceries has a correlation of 0.70 with Transport, 0.68 with Eating_Out, and 0.69 with Miscellaneous. This implies that people who can save in one category tend to show a similar potential to others. This observation hints at higher-level financial behavior groups: people who manage spending of discretionary nature well in one category typically display capacity in others as well. Interestingly, Potential_Savings_Education shows the overall lowest correlation compared to other categories, such as Eating_Out (0.49) and Healthcare (0.50), which may reflect that it is more of the planned or non-discretionary type of expense that varies much on the basis of life stage or structure of the household. The declining correlation of educational savings potential with other categories constitutes an argument that educational expenses are less substitutable and less behavior-driven, as opposed to lifestyle-related expenses. The utility of this heatmap lies in its importance for dimensionality reduction and feature selection for predictive modeling, with the majority of features being highly correlated and probably suffering from multicollinearity. Hence, could be resolved by PCA or regularization to avoid overfitting of the model. Clustering of these variables can be used in the definition of composite indices of financial behavior or in the segmentation of user profiles, both of which are extremely useful in applications such as personalized financial planning or targeted intervention strategies.

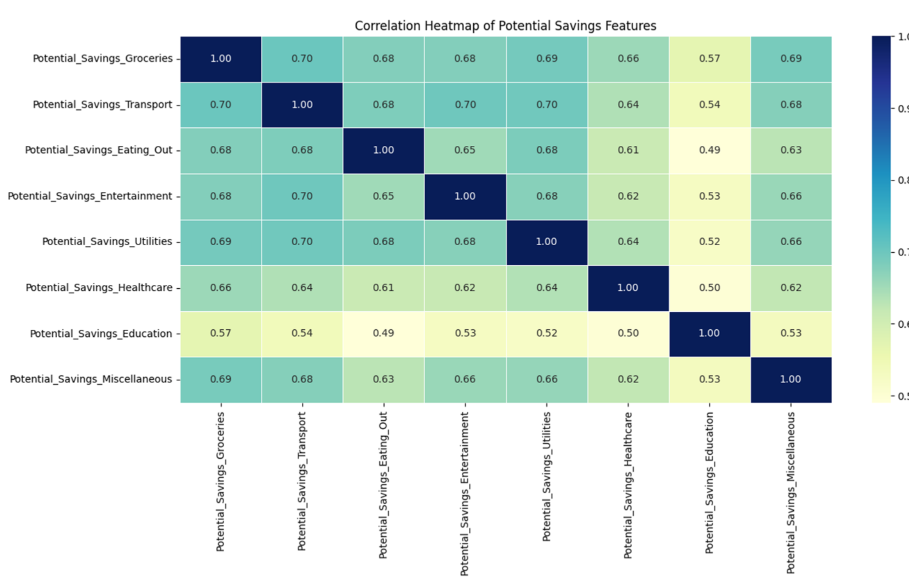

Fig. 5 Correlation heatmap of potential saving features

A bar chart titled "Average Monthly Expenses by Occupation" in Fig. 6 compares average monthly spending for Groceries, Transport, and Healthcare for a range of occupational groups: Professional, Retired, Self-Employed, and Student. The trend in the data remains consistent throughout the chart; groceries have the highest average monthly expenses for all occupational categories, more than ₹5,000. This suggests that food-related expenses mostly take precedence as a financial consideration, regardless of employment situation. Once again, transportation comes next, and then healthcare maintains a relatively constant level for all groups at ₹1,600–₹1,700. This presentational view shows that there seems to be little variation among the different occupational groups in the expenditures for these essentials, which might insinuate that lifestyle expenses are more or less the same due to possibly unified pricing or almost equivalent access levels.

The significance of the graph lies in its implications for personalized financial planning and targeted savings interventions. Social spending patterns, depending upon occupation, may, therefore, also be largely similar, since similar social spending baskets, policymakers, app developers, or financial institutions could target generic budgeting or saving advice that applies to almost all end-buyer segments. The graph also serves as a starting point for the analysis of spending efficiency or potential savings with reference to occupational profiles.

3.3 Feature scaling

To scale the important features, the TabNet Feature Importance illustrated in Fig. 7 indicates the relative contribution of each feature toward the model's prediction of financial categorization or stability [18]. Debt_Ratio was the predominant feature, accounting for almost 50% of the decision weight of the model. The dominant position of Debt_Ratio clearly shows that the amount of income paid for loans is perhaps the most important indicator of financial well-being or risk. Potential_Savings_Miscellaneous and Savings Ratio are key features, indicating that miscellaneous cash spending and the proportion of income saved are strong predictors of financial wellbeing. Other features such as Desired_Savings_Percentage, Insurance, Occupation, and Potential_Savings_Utilities exhibit moderate importance, indicating that while they impact the outcome, this influence can be situational and somewhat indirect. Surprisingly, basic variables such as

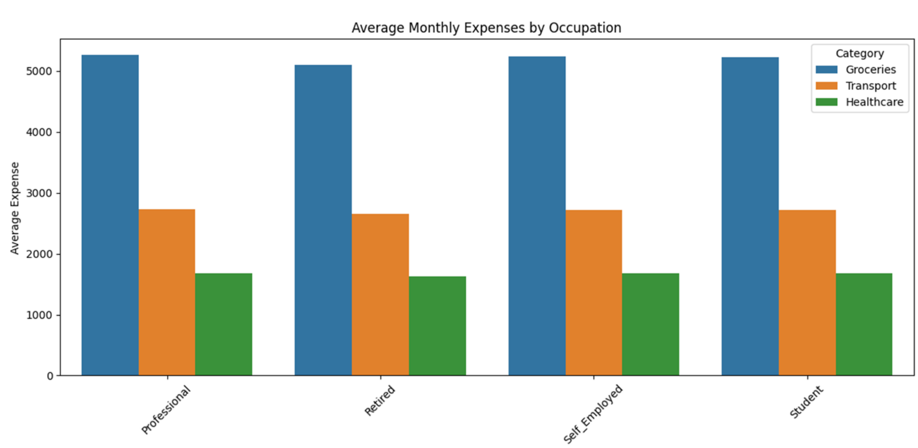

Fig. 6 Average monthly expenses by occupation

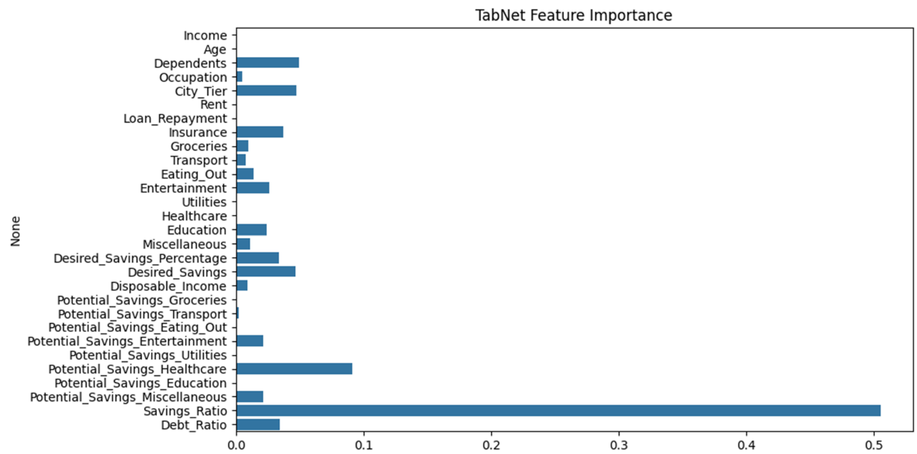

Fig. 7 Feature importance using TabNet

Income, Disposable_Income, and Age show minimal importance, which could arise with some of their information being indirectly captured or normalized through constructed ratios, such as the Debt Ratio or Savings Ratio. Similarly, the standalone categories of Groceries, Transport, and Healthcare show low predictive power, reinforcing that the model finds more value in composite financial behavior (in terms of ratios and potential savings) than in raw monetary values.

The importance of this section lies in guiding the selection of features, intervention strategies, and model interpretability. For instance, financial advisory systems may focus on monitoring and reducing users' debt ratios and increasing savings in discretionary areas at the model level rather than simply increasing users' income. It also validates feature engineering (e.g., ratio-based variables) that help with the model as well as real-world insight.

3.4 Classifiers for analysing the pattern of finance

To model and classify patterns from the Indian Personal Finance and Spending Habits dataset, we adopted a sequence of advanced deep-learning architectures, each tailored to the unique characteristics of the data. The dataset, consisting of both categorical and numerical features such as income, investment preference, credit card usage, and spending tendencies, was first preprocessed and transformed into a normalized numerical form $X = \{x_1, x_2, \dots, x_n\}$, where each $x_i \in R^d$ represents a d-dimensional feature vector for the i-th individual. We began by applying Deep Neural Networks (DNNs) and Fully Connected Neural Networks (FCNNs), which utilized multiple hidden layers with ReLU activation $f(x) = \max(0, x)$ to learn complex, nonlinear feature interactions [20]. To capture localized feature dependencies, we reshaped the tabular input into pseudo-sequences and applied Convolutional Neural Networks (CNNs), where the convolution operation was defined as $h_i = \sigma\left(\sum_{j=1}^{k} w_j \cdot x_{i+j-1} + b\right)$, allowing the model to learn spatial hierarchies among grouped financial indicators. Given the potential temporal influence on decision-making (e.g., prior savings affecting current investments), we integrated Recurrent Neural Networks (RNNs) and advanced them using Gated Recurrent Units (GRUs) and Bidirectional Long Short-Term Memory networks (BiLSTMs), where the cell state updates are governed by gates such as $f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right)$, enabling effective modeling of sequential financial

```
Input:
   D ⟵ Financial dataset with numerical and categorical features
   E ⟵ Number of training epochs
   B ⟵ Batch size
   α ⟵ Learning rate

Procedure:
1. Preprocessing:
  a. One-hot encode all categorical variables in D.
  b. Normalize all numerical variables using Min-Max scaling.
  c. Concatenate encoded categorical and normalized numerical features to form x_i ∈ ℝ^d.
  d. Reshape x_i into a 2D tensor suitable for 1D convolution: x_i ∈ ℝ^{1×d}.

2. Wide Component Construction:
  a. Define feature crosses from selected raw input features.
  b. Compute wide output using linear transformation:
    y_wide = W_wideᵀ · x_i + b_wide

3. Deep CNN Component Construction:
  a. Apply 1D convolution to x_i:
    h_conv = ReLU(Conv1D(x_i, filters, kernel_size) + b_conv)
  b. Apply MaxPooling to reduce dimensionality:
    h_pool = MaxPool1D(h_conv)
  c. Flatten the pooled output:
    h_flat = Flatten(h_pool)
  d. Pass h_flat through fully connected dense layer(s):
    y_deep = Dense(h_flat)

4. Merge Wide and Deep Components:
  a. Combine outputs:
    y_total = σ(y_wide + y_deep)
    where σ is the sigmoid activation for binary classification

5. Training:
  a. For each epoch from 1 to E:
    i. Divide data into mini-batches of size B
    ii. For each batch:
      - Perform forward propagation through both wide and deep branches
      - Compute loss using binary cross-entropy:
        L = −[y · log(y_total) + (1−y) · log(1−y_total)]
      - Backpropagate gradients
      - Update model parameters using Adam optimizer with learning rate α

6. Return the trained model

End Procedure

Output:
   Trained Wide & Deep + CNN model, Financial Status Classification
```

Algorithm 1 Feature importance using TabNet

behaviors. Recognizing that some survey responses might exhibit long-range dependencies, BiLSTM offers a bidirectional context across time [3]. To simultaneously benefit from memorization and generalization, we implemented at the Wide & Deep model, where the prediction was computed as $y = \sigma\left(W_{\text{wide}}^{T}x + W_{\text{deep}}^{T}\varphi\ (x)\right)$, where x represents the raw features and $\varphi\ (x)$ is the transformation learned by the DNN. We extended this paradigm by combining the wide component with sequential layers such as BiLSTM, CNN, GRU, and Attention, forming Wide & Deep + BiLSTM, Wide & Deep + CNN, and Wide & Deep + Attention models, each designed to simultaneously learn low-level interactions and temporal abstractions [22]. Furthermore, we incorporated autoencoders to compress the high-dimensional feature space into a latent representation $z = f_{\text{enc}}(x)$, followed by decoding $\widehat{x} = f_{\text{dec}}(z)$, using the reconstruction loss $L_{\text{rec}} = \| x - \widehat{x}\|^{2}$ to guide representation learning. These encodings are then fed into classifiers or RNNs for downstream classification. Hybrid architectures, such

```
Input:
  D ← Financial dataset with behavioral features
  E ← Number of training epochs
  B ← Batch size
  α ← Learning rate

Procedure:
1. Preprocessing:
  a. One-hot encode categorical variables.
  b. Normalize numerical variables using Z-score or Min-Max normalization.
  c. Organize features into pseudo-sequential form (if not temporal):
    X_i = [x_i¹, x_i², ..., x_i^T], where T is fixed sequence length

2. Wide Component Construction:
  a. Identify feature interactions and generate cross-product terms.
  b. Compute wide output:
    y_wide = W_wideᵀ · x_i + b_wide

3. Deep BiLSTM Component Construction:
  a. Feed X_i into Bidirectional LSTM layer:
    H = BiLSTM(X_i)
  b. Concatenate last hidden states from both directions:
    H_concat = [h_T^forward ; h_T^backward]
  c. Pass concatenated output through a dense layer:
    y_deep = Dense(H_concat)

4. Merge Wide and Deep Components:
  a. Combine outputs:
    y_total = Softmax(y_wide + y_deep) for multi-class
          or
    y_total = Sigmoid(y_wide + y_deep) for binary

5. Training:
  a. For each epoch from 1 to E:
    i. Shuffle and divide D into mini-batches of size B
    ii. For each batch:
      - Execute forward pass through both components
      - Compute cross-entropy loss:
        L = −Σ [y_c · log(y_total_c)] for c = 1 to C classes
      - Apply backpropagation
      - Update weights using Adam optimizer with learning rate α

6. Return the trained model

End Procedure

Output:
  Trained Wide & Deep + BiLSTM model, Financial Status Classification
```

Algorithm 2 Wide & deep + BiLSTM for financial behavior classification

as CNN + LSTM and CNN + MLP, were used to first extract patterns using convolutional layers and subsequently model sequential or dense interactions through LSTM or MLP units. To enhance attention toward critical features (e.g., those influencing debt or savings behavior), we employed Attention-based CNNs and Wide & Deep + Attention mechanisms, where the attention score was computed as $\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}$ with $e_i = v^T \tanh(Wx_i + b)$, dynamically weighting the key features in the decision process. Collectively, this ensemble of models enables comprehensive representation learning from heterogeneous personal finance attributes, capturing both static and dynamic behavioral patterns to yield robust classification outcomes [19].

A hybrid architecture combining a Wide & Deep learning framework with a Convolutional Neural Network (CNN) is shown in Algorithm 1 for classifying individuals based on their financial behavior. In this manner, a model can capture the interactions of low-order features in parallel with computations to extracting patterns of higher order through one-dimensional convolutions applied to appropriately reshaped financial-feature vectors. Because the data are structured in tabular format, wherein the absence of temporal ordering is associated with embedded local correlations, this architecture is ideal for such cases. This algorithm provides the details for pre-processing, architectural flow, and optimization to be used for training the Wide & Deep + CNN model on multi-financial-classification tasks [26].

Algorithm 2 provides an outline of building and training the hybrid Wide & Deep model using the BiLSTM network for financial behavior classification. This model uses the generalization and memorization of the third layer. Hence, the wide layer is considered for memorization and pooling the specification of feature interactions, and the deep BiLSTM layer is considered for generalization by militating sequential or pseudo-sequential patterns inherent in behavioral financial data. The algorithm starts with structured preprocessing, with the distortion of wide and deep components in parallel, which are later joined and optimized using a supervised learning procedure with Adam optimizer [6].

4 Result and analysis

This section presents the empirical outcomes obtained by applying various deep learning classifiers to the Indian personal finance and spending habit dataset.

In Table 2, the training and validation performance metrics reveal several key insights into model behavior on the Indian personal finance dataset. First, models such as Wide & Deep + CNN, Wide & Deep + BiLSTM, and BiLSTM not only achieved the highest validation accuracies (≥ 0.988), but also maintained very low validation losses (~ 0.025–0.026), indicating their strong generalization ability. This suggests that models capable of both memorization (via wide components) and sequence modeling (via BiLSTM/CNN) are particularly effective at capturing the complex interdependencies among features, such as income, expense categories, city tier, and savings behavior. The minimal gap between the training and validation metrics in these models indicated stable learning and a low risk of overfitting.

Table 2 Performance comparison of deep learning models on financial data

Model	Train accuracy	Train loss	Val accuracy	Val loss
CNN	0.8788	0.3006	0.8687	0.3182
RNN	0.9773	0.0525	0.9791	0.0440
DNN	0.9845	0.0379	0.9766	0.0511
BiLSTM	0.9851	0.0324	0.9884	0.0256
GRU	0.9905	0.0219	0.9859	0.0255
Wide & Deep	0.9916	0.0283	0.9831	0.0456
Autoencoder + Classifier	0.9827	0.0457	0.9803	0.0438
FCNN	0.9743	0.0602	0.9762	0.0683
Attention-CNN	0.8835	0.2798	0.8697	0.2901
Residual MLP	0.9798	0.0480	0.9578	0.1066
Hybrid CNN + MLP	0.7165	− 94653360.0	0.7025	− 103436192.0
Hybrid CNN + LSTM	0.8717	0.3232	0.8544	0.3617
Hybrid CNN + GRU	0.8783	0.2975	0.8622	0.3315
Hybrid RNN + BiLSTM	0.9834	0.0403	0.9781	0.0464
Autoencoder + RNN	0.9643	0.0856	0.9584	0.0941
Wide & Deep + BiLSTM	0.9880	0.0305	0.9900	0.0261
Wide & Deep + CNN	0.9937	0.0268	0.9944	0.0259
Wide & Deep + RNN	0.9772	0.0571	0.9844	0.0441
Wide & Deep + Attention	0.9879	0.0354	0.9897	0.0368

By contrast, models such as CNN, Hybrid CNN + GRU, and Attention-CNN show larger validation losses (~ 0.29–0.33) and slightly lower accuracies (~ 0.86), reflecting their inability to fully capture high-dimensional financial patterns. This limitation likely stems from their reliance on local feature extraction without recurrent memory or dynamic feature weighting, which are essential given the dataset's multi-feature correlations (e.g., income vs. groceries and, income vs. savings).

Moreover, the Hybrid CNN + MLP model is severely unstable, showing negative loss values in the order of millions, indicating a critical training failure, possibly due to incompatible layer integration or numerical overflow. Interestingly, even simpler models, such as RNN, FCNN, and Autoencoder + Classifier, maintain robust validation accuracy (> 0.95) and low loss, highlighting that even without sophisticated hybridization, models that can extract latent feature relationships perform well because of the structured and patterned nature of the dataset.

In summary, the performance table clearly supports the conclusion that models with architectural components for hierarchical abstraction (deep layers), temporal/sequential modeling (RNN, BiLSTM), and feature attention or selection (Wide & Deep, TabNet) are the most suitable for this type of financial behavior data.

Additionally, the learning curves and confusion matrix of the best-performing models are also presented in Fig. 8, where both hybrid models show a good fit of models in training and validation.

The overall performance of the models on the Indian Personal Finance and Spending Habits dataset reflects the nature of the data, which are highly structured, multivariate, and deeply interdependent, as shown in Table 3. Models such as BiLSTM, GRU, and hybrid architectures such as Wide & Deep + CNN, Wide & Deep + BiLSTM, and TabNet achieve perfect or near-perfect scores (F1 ≈ 0.99–1.00), showcasing their ability to model complex feature interactions, temporal dependencies, and hierarchical relationships. These models excel because the dataset contains patterns such as strong positive correlations between income and multiple expense categories (e.g., rent, groceries, transport) and between income and savings, which are best captured by architectures capable of nonlinear representation learning and memory mechanisms. For instance, the BiLSTM and GRU models retain long-range dependencies and are robust to feature skewness and class imbalance, explaining their 1.00 precision, recall, and F1 scores.

Deep neural models such as DNN, FCNN, and Residual MLP also perform competitively (F1 ≈ 0.94–0.97) owing to their capacity to learn hierarchical abstractions from numeric input features. By compressing high-dimensional data into efficient latent representations, autoencoder-based models also preserve essential variance while reducing redundancy, resulting in F1 scores of up to 0.98. The Wide & Deep model and its enhanced hybrids benefit from combining memorization (wide component) with generalization (deep component), particularly in a domain such as personal finance where both rare and frequent patterns coexist.

On the other hand, CNN-based architectures, including Hybrid CNN + GRU/LSTM and Attention-CNN, show lower performance (F1 ≈ 0.84–0.86), and in the case of Hybrid CNN + MLP, they drastically underperform (F1 ≈ 0.28). This underperformance likely stems from CNNs' limited capacity of CNNs to model long-range dependencies in non-spatial tabular data, making them less suited for financial datasets that require an understanding of contextual interactions across features. Moreover, models such as

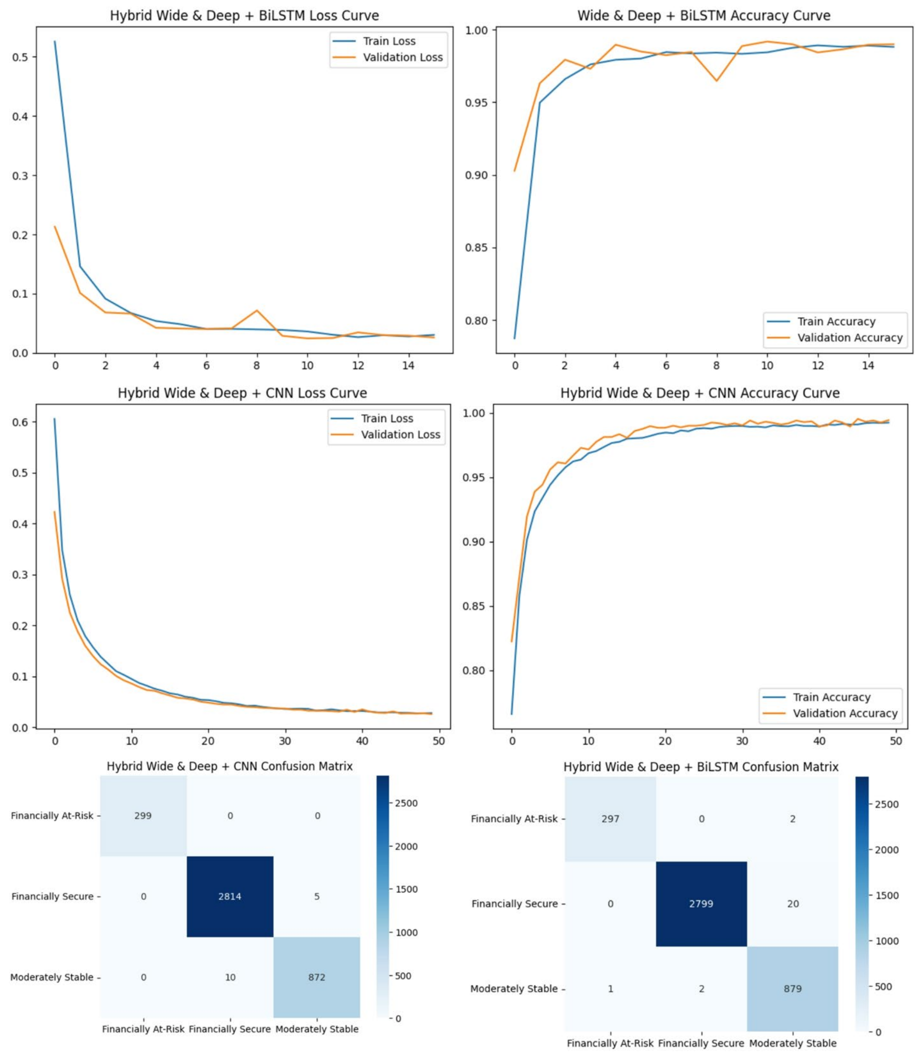

Fig. 8 Analysis of the best performing models

Table 3 Examination of the trained classifiers for analysing financial dataset

Model	Precision	Recall	F1-Score	Model	Precision	Recall	F1-Score
CNN	0.90	0.84	0.86	FCNN	0.97	0.97	0.97
RNN	0.98	0.99	0.99	Attention-CNN	0.92	0.84	0.86
DNN	0.98	0.97	0.97	Residual MLP	0.97	0.92	0.94
BiLSTM	1.00	1.00	1.00	TabNet	0.99	0.99	0.99
GRU	1.00	1.00	1.00	Hybrid CNN + MLP	0.23	0.33	0.28
Wide & Deep	0.99	0.98	0.98	Hybrid CNN + LSTM	0.92	0.81	0.84
Autoencoder + Classifier	0.97	0.98	0.98	Hybrid CNN + GRU	0.90	0.82	0.85
Hybrid RNN + BiLSTM	0.98	0.98	0.98	Wide & Deep + CNN	1.00	1.00	1.00
Autoencoder + RNN	0.94	0.96	0.95	Wide & Deep + RNN	0.99	0.99	0.99
Wide & Deep + BiLSTM	0.99	0.99	0.99	Wide & Deep + Attention	0.99	0.99	0.99

Attention-CNN and Hybrid CNN + LSTM may suffer from overfitting or ineffective attention calibration, particularly when feature importance is unevenly distributed.

Likewise, Table 4 contains precision, recall, and F1-score data for the applied deep learning architectures evaluated in the financial health categories of Financially At-Risk, Financially Secure, and Moderately Stable.

The outstanding performance of advanced sequence-based models such as BiLSTM, GRU, and various hybrid architectures (e.g., Wide & Deep + CNN, Wide & Deep + BiLSTM) on this dataset likely stems from the rich multivariate structure and strong inter-feature correlations of the dataset. Seq-models, such as RNNs, LSTMs, and GRUs, are adept at capturing complex, context-dependent patterns with sequential or time-related features, allowing them to tease apart subtle distinctions among classes. Indeed, RNN achieved nearly perfect metrics across categories, whereas BiLSTM and GRU reached near-ideal precision, recall, and F1 across all classes. These models can effectively model interactions (e.g., how income interacts with multiple expense categories and demographics) and are robust to the noise and outliers common in skewed financial data. In contrast, simpler CNNs and hybrid shallow-deep models may struggle to fully represent these multifaceted relationships. For example, CNNs showed drops in F1 for "Moderately Stable" (~ 0.67), likely because convolutional filters alone do not adequately model feature interactions over all input vectors. Similarly, models such as Attention-CNN or Hybrid CNN+LSTM/GRU improved stability moderately but still lagged behind full RNN architectures. Autoencoders with classifiers also scored well (~ 0.97 F1) because they adaptively compressed the high-dimensional skewed input features before classification, effectively performing manifold learning that benefits downstream classification. TabNet, with its sequential attention mechanism over tabular data, performed extremely well (F1 ~ 0.99) by dynamically selecting and weighting the relevant feature subsets.

Similarly, Fig. 9 illustrates the ROC-AUC scores of various deep learning and hybrid models developed for financial health prediction. ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) is a critical metric that evaluates a model's ability to distinguish between different classes and in this case, individuals categorized as *Financially Secure*, *Moderately Stable*, or *Financially At-Risk*. Models such as BiLSTM, GRU, and the Wide & Deep + CNN hybrid achieved a perfect ROC-AUC score of 1.0000, indicating flawless discrimination across financial well-being categories. This underscores their exceptional reliability in real-world deployment in personal finance advisory systems. Other top-performing models include TabNet, RNN, Wide & Deep + BiLSTM, Autoencoder + Classifier, and Wide & Deep + RNN, all exceeding a score of 0.999, reflecting near-perfect classification ability. Traditional architectures such as CNN and hybrids such as CNN + LSTM and CNN + GRU showed relatively lower but still commendable ROC-AUC values, ranging from 0.9286 to 0.9428. This finding suggests room for improvement, particularly in handling the nonlinearity and complexity of financial behavior patterns. The graph highlights a clear trend: hybrid models leveraging architectural synergies such as Wide & Deep sequence models) outperform standalone models. This aligns with contemporary machine learning theory, which posits that hybrid architectures can capture both static and sequential patterns in complex datasets crucial for financial well-being assessments that depend on both categorical traits and spending behaviors. Overall, the graph provides compelling evidence for the superiority of hybrid deep learning architectures in achieving robust financial classification and actionable insights for researchers and practitioners aiming to build scalable and interpretable financial decision support systems.

Furthermore, Table 5 compares existing work on segmentation, systemic risks, and stress testing with the proposed framework that uses large-scale real-world data and hybrid, highly synergistic deep learning models to segregate individual clients as either

Table 4 Class-wise performance metrics of deep learning models for financial health classification

Model	Class	Precision	Recall	F1-Score
CNN	Financially At-Risk	0.99	1.00	0.99
	Financially Secure	0.88	0.97	0.92
	Moderately Stable	0.85	0.56	0.67
RNN	Financially At-Risk	0.98	1.00	0.99
	Financially Secure	1.00	0.99	1.00
	Moderately Stable	0.98	0.98	0.98
DNN	Financially At-Risk	0.97	0.95	0.96
	Financially Secure	0.99	1.00	0.99
	Moderately Stable	0.98	0.95	0.96
BiLSTM	Financially At-Risk	1.00	1.00	1.00
	Financially Secure	1.00	1.00	1.00
	Moderately Stable	1.00	0.99	0.99
GRU	Financially At-Risk	1.00	1.00	1.00
	Financially Secure	1.00	1.00	1.00
	Moderately Stable	1.00	0.99	0.99
Wide & Deep	Financially At-Risk	0.99	0.96	0.98
	Financially Secure	0.99	1.00	1.00
	Moderately Stable	0.98	0.98	0.98
Autoencoder + Classifier	Financially At-Risk	0.95	0.99	0.97
	Financially Secure	1.00	0.99	0.99
	Moderately Stable	0.97	0.97	0.97
FCNN	Financially At-Risk	0.94	0.98	0.96
	Financially Secure	0.99	0.99	0.99
	Moderately Stable	0.97	0.94	0.96
Attention-CNN	Financially At-Risk	0.97	0.99	0.98
	Financially Secure	0.87	0.99	0.93
	Moderately Stable	0.92	0.53	0.67
Residual MLP	Financially At-Risk	1.00	0.79	0.88
	Financially Secure	0.99	0.99	0.99
	Moderately Stable	0.92	0.97	0.94
TabNet	Financially At-Risk	0.99	0.99	0.99
	Financially Secure	1.00	1.00	1.00
	Moderately Stable	0.98	1.00	0.99
Hybrid CNN + LSTM	Financially At-Risk	0.98	0.97	0.98
	Financially Secure	0.86	0.99	0.92
	Moderately Stable	0.94	0.46	0.62
Hybrid CNN + GRU	Financially At-Risk	0.98	0.97	0.98
	Financially Secure	0.87	0.97	0.92
	Moderately Stable	0.85	0.53	0.65
Hybrid RNN + BiLSTM	Financially At-Risk	0.95	0.99	0.97
	Financially Secure	0.99	1.00	0.99
	Moderately Stable	0.98	0.96	0.97
Autoencoder + RNN	Financially At-Risk	0.90	0.98	0.94
	Financially Secure	0.99	0.98	0.98
	Moderately Stable	0.93	0.92	0.93
Wide & Deep + BiLSTM	Financially At-Risk	1.00	0.99	0.99
	Financially Secure	1.00	0.99	1.00
	Moderately Stable	0.98	1.00	0.99
Wide & Deep + CNN	Financially At-Risk	1.00	1.00	1.00
	Financially Secure	1.00	1.00	1.00
	Moderately Stable	0.99	0.99	0.99

Table 4 (continued)

Model	Class	Precision	Recall	F1-Score
Wide & Deep + RNN	Financially At-Risk	0.98	1.00	0.99
	Financially Secure	0.99	1.00	0.99
	Moderately Stable	1.00	0.96	0.98
Wide & Deep + Attention	Financially At-Risk	0.98	1.00	0.99
	Financially Secure	1.00	1.00	1.00
	Moderately Stable	0.98	0.99	0.99

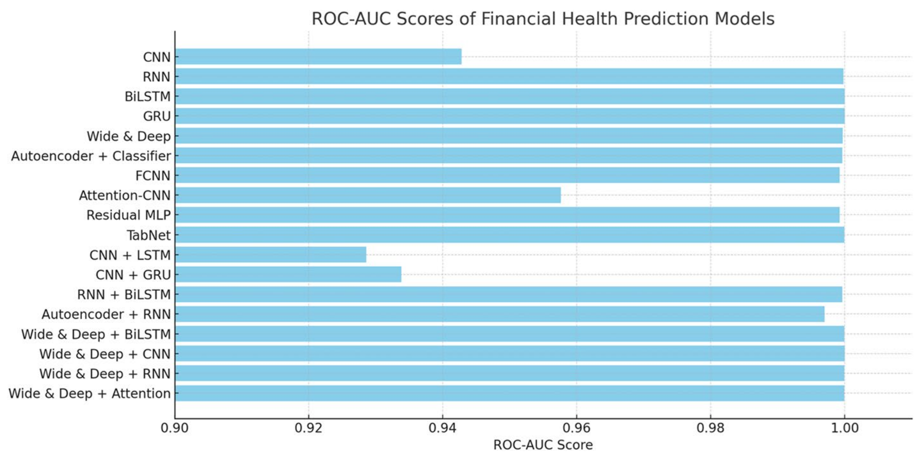

Fig. 9 ROC-AUC score of financial health prediction models

Financially Secure, Moderately Stable, or Financially At-Risk. It is superior to existing approaches in all key performance metrics, allowing for greater interpretability and a consequential deployment scope for integration into financial advisory and wellness platforms.

5 Conclusion, limitations, and future scope

This study proposes a novel hybrid deep learning framework for carrying out robust financial profiling of persons into Financially Secure, Moderately Stable, and Financially At-Risk. Using a large-scale dataset of 20,000 Indian individuals with income, expenses, savings, and demographic features, this study presents the higher efficacy of advanced hybrid models, especially Wide & Deep + CNN and Wide & Deep + BiLSTM. These models achieved almost perfect classification metrics (Validation Accuracy: 99.44%, F1-Score: 1.00, ROC-AUC: 1.0000), thus outperforming regular CNNs and less-complex hybrid models. Furthermore, the pipeline uses TabNet to improve explainability, which supports transparency during decisions, thus moving toward real-world applications such as finance advice, stress detection, and aided budgeting. However, there are some limitations to this work despite its impressive performance. A primary limitation is that, while the dataset has considerable variables pertaining to structure, it lacks temporal sequences or longitudinal financial behaviors, which are important in both real-time financial forecasts and modeling trajectories of stress. Second, the data remained geographically and culturally confined to Indian individuals, possibly limiting the generalizability of the models to other populations with different financial habits or economic systems. Furthermore, some hybrid models (e.g., CNN + MLP) were found to be unstable because of architectural incompatibilities, suggesting that rigorous integration should

Table 5 Comparative analysis of existing techniques vs. the proposed hybrid deep learning approach for financial health classification

Dimension	Existing techniques	Proposed technique
Research objective	Focused on segmentation (Researcher 1), welfare impact of banking crises (Researcher 2), systemic AI risks (Researcher 3), financial stress testing (Researchers 5 & 6), and early bankruptcy prediction (Researchers 7 & 8)	Accurately classify individuals into *Financially Secure*, *Moderately Stable*, and *Financially At-Risk* using hybrid deep learning models
Dataset	Small-scale or synthetic datasets: 1,874 survey responses (Researcher 1), simulation data (Researcher 2), or stock indicators (Researcher 9)	Real-world, large-scale dataset of 20,000 Indian individuals with detailed income, expenditure, savings, and demographic data
Input features	Survey-based (Researcher 1), macroeconomic indicators (Researcher 2), stock returns and liquidity ratios (Researcher 9)	Multivariate features: income, savings ratio, debt ratio, monthly expenses across categories, city tier, occupation, etc.
Modeling techniques	Clustering (Researcher 1), Agent-Based + SVM (Researcher 2), Gradient Boosting + Optimizers (Researcher 8), CNN + LSTM (Researcher 5)	15 + deep learning models including CNN, RNN, BiLSTM, GRU, DNN, FCNN, TabNet, Autoencoder-based classifiers, and Wide & Deep hybrids
Hybridization	CNN + LSTM/GRU (Researcher 5), CNN + Text Analysis (Researcher 6), optimization-driven DL (Researcher 7)	Wide & Deep integrated with CNN, BiLSTM, RNN, and Attention mechanisms for joint memorization and generalization
Key metrics	Accuracy: ~95.8% (AWOA-DL, Researcher 7), Precision: 0.980 (HGSO, Researcher 8), moderate ROC-AUC (CNN-based models)	Validation Accuracy: 99.44%; ROC-AUC: 1.0000; F1-Score: 1.00 (GRU, BiLSTM, Wide & Deep + CNN)
Model output	Binary segmentation (Researcher 1), risk classification (Researchers 5–8)	Multi-class classification with granular breakdown of three financial health levels
Interpretability	Often criticized as black-box (Researcher 4); limited transparency	Uses TabNet, feature engineering (Savings/Debt Ratio) to improve model explainability
Systemic insights	Highlights macroeconomic resilience and regulatory blind spots (Researcher 3)	Supports individual-level predictions with potential for integration into organizational wellness systems
Deployment readiness	Conceptual or research-stage tools; not linked to real-time systems	Designed for real-time financial profiling, robo-advisory, and employee stress mitigation platforms

be designed. While interpretability remains aided by feature selection, some stakeholders unfamiliar with AI techniques find it a complex challenge. Several promising avenues for future research can be explored. Real-time financial transaction data could be included in the pipeline, and metrics from behavioral psychology could be integrated to make predictions more dynamic and personalized. Enriching the dataset to contain individuals from different geographies and socioeconomic backgrounds will put the model on a more robust footing. The predictive pipeline can also be implemented in a mobile application or a financial chatbot to connect academic innovations with their practical deployment Finally, in sensitive areas like finance, AI deployment should be considered with aspects such as fairness, mitigation of biases, and transparency.

Author contributions
[AU]: Conception and design of the study, supervision of the research process, and critical revision of the manuscript for intellectual content.AS]: Data curation, model development, and analysis and interpretation of results.[YA]: Drafting of the manuscript, literature review, and visualization of findings.[AS & BK]: Editing, proofreading, and ensuring methodological rigor.All authors were involved in critically revising the paper for its intellectual content, approved the final version to be published, and agreed to be accountable for all aspects of the work, ensuring that questions related to accuracy or integrity were appropriately investigated and resolved.

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Data availability
The data that support the findings of this study are openly available in [Kaggle.com]at [https://www.kaggle.com/datasets/shriyashjagtap/indian-personal-finance-and-spending- habits](https:/www.kaggle.com/datasets/shriyashjagtap/indian-personal-finance-and-spending-%20habits), [13].

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Informed consent
This study did not involves human participants.

Human and animals participants
This research did not involve any studies with human participants or animals performed by any of the authors.

Competing interests
The authors declare no competing interests.

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